Biopharmaceutical Operations

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Biopharmaceutical Operations (A33642)

Short Title:	Biopharm Ops		
Faculty	Science and Computing		
Department	Science		
Cluster	Pharmaceutical Science		
Author	AHAYDEN		
Credits	10	**Level:**	Advanced

Description of Module / Aims

This module will provide the learner with an in depth understanding of the key operations involved in the manufacture of a Biopharmaceutical product. Key areas covered in the module are Upstream and Downstream Processing, Biopharmaceutical stabilisation and formulation, Lyophilisation of Biopharmaceuticals, packaging, sterilisation and containment and critical manufacturing operations.

Programmes

`PHAR-0024`	BSc (Hons) in Good Manufacturing Practice and Technology (`WD_SGMPT_B`)	4 / 2 / M

Indicative Content

- Upstream Processing: Media selection and Scale-up issues. Critical Processing parameters for cell growth. Process monitoring. The use, type, design and operation of Bioreactors.
- Downstream Processing: Harvesting, Cell disruption -- Intracellular/Extracellular, centrifugation, extraction, filtration and purification techniques, membranes, chromatography systems.
- Lyophilisation: Freeze-drying processing. Scale-up and stabilisation of biopharmaceutical compounds.
- Analytical techniques for development and monitoring of lyophilised Biopharmaceutical formulations.
- Container integrity and Packaging of Biopharmaceuticals.
- Clean room containment and sterilisation, CIP/SIP. Layout and Design. HVAC. Environmental monitoring

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate a clear understanding the upstream and downstream processing of biopharmaceuticals and critical processing parameters involved.
2. Determine key operations involved in lyophilisation of Biopharmaceutical products.
3. Characterise and evaluate finished biopharmaceutical products.
4. Appraise and review formulation activities for biopharmaceutical compounds.
5. Interpret and evaluate techniques used to ensure sterility in Biopharmaceutical production.

Learning and Teaching Methods

- Lectures.
- Industry based learning and Independent research and reading

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Assignment	30%	1,2,3,4,5

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of Biopharmaceutical production or Lyophilisation. Lack of competence with regard to associated assignments.

40%-49%: Will be able to show a limited understanding of the fundamentals of Biopharmaceutical Operations, but may have some difficulty with applying this theory. Will display some ability with regard to basic overviews of processing.

50%-59%: As well as the above, the candidate will be able to show a reasonable understanding of the complexities of Biopharmaceutical Processing. Will display reasonable competence with regard to associated unit processes and underlying principles.

60%-69%: In addition, the candidate will demonstrate a more detailed knowledge of the Biopharmaceutical operations, the associated equipment and the sterilised environment.

70%-100%: All of the above to an excellent level. In addition, the candidate will be able to demonstrate comprehensive of all aspects of Biopharmaceutical Operations, critical processing parameters and effects on product quality as well as product testing and sterile manufacturing capabilities.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Essential Material

- Franks, F. _Freeze-Drying of Pharmaceuticals and Biopharmaceuticals: Principles and Practice._ . 1st. Cambridge U.K..: Royal Society of Chemistry, 2007.
- Gilleskie , G. and C. Rutter. _Biopharmaceutical Manufacturing: Principles, Processes, and Practices_ . North Carolina USA: De Gruyter, 2021.

Supplementary Material

- Whyte, W. _Cleanroom Technology: Fundamentals of Design, Testing and Operation,_. 2nd. U.K.: Wiley, 2010.

Requested Resources

- Lecture Room: Loose Seated